

JAVA ITERATOR

DATE:06/08/26

1.

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS
Q1
A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.
10 marks
Q2
Remove Even Numbers Using Iterator
10 marks
Q3
Product Price Analysis Using Iterator
10 marks
1/3 answered

Q1
A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.
10 marks
A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.
Write a Java program to:
Create an ArrayList of student names.
Read 8 student names from the user.
Traverse the list using an Iterator.
Display each student name on a separate line.
Display the total number of students stored in the list.

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.Scanner;
4 class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<String> students = new ArrayList<>();
8         System.out.println("Enter 10 student names:");
9         for (int i = 0; i < 10; i++) {
10             students.add(sc.nextLine());
11         }
12         System.out.println("\nStudent Names:");
13         Iterator<String> it = students.iterator();
14         while (it.hasNext()) {
15             System.out.println(it.next());
16         }
17         System.out.println("\nTotal Students = " + students.size());
18         sc.close();
19     }
20 }
```

Input
FGDPS
EGR
sdF
vds
vds

Output
FGDPS
EGR sdF
sdF
vds
Total Students = 10

Run

Save

2.

**Programiz**
Online Java Compiler

Programiz PRO

Main.java

```
1- import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.Scanner;
4
5- class Main {
6-     public static void main(String[] args) {
7-
8-         Scanner sc = new Scanner(System.in);
9-
10-         ArrayList<Integer> emp = new ArrayList<>();
11-
12-         System.out.println("Enter 10 Employee ID numbers:");
13-
14-         for (int i = 0; i < 10; i++) {
15-             emp.add(sc.nextInt());
16-         }
17-
18-         Iterator<Integer> it = emp.iterator();
19-
20-         while (it.hasNext()) {
21-             int id = it.next();
22-
23-             if (id % 2 == 0) {
24-                 it.remove();
25-             }
26-         }
27-
28-         System.out.println("Employee IDs after removing even numbers:");
29-
30-         for (int id : emp) {
31-             System.out.print(id + " ");
32-         }
33-
34-         sc.close();
35-     }
36 }
```

Output

Enter 10 Employee ID numbers:
12
45
5
78
36
4
46
14
47
3
Employee IDs after removing even numbers:
45 5 47 3
=== Code Execution Successful ===

3.



SIMATS Online Programming Lab
JAVA PROGRAMMING


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Q2
Remove Even Numbers Using Iterator
10 marks

Q3
Product Price Analysis Using Iterator
10 marks

3/3 answered

Read the prices of 10 products from the user.

Traverse the list using an Iterator.

Calculate and display:

- Total price of all products.
- Average product price.
- Highest product price.
- Display all product prices using the Iterator.

Your Code

```

1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.Scanner;
4 class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<Double> price = new ArrayList<>();
8         System.out.println("Enter 10 product prices:");
9         for (int i = 0; i < 10; i++) {
10             price.add(sc.nextDouble());
11         }
12         Iterator<Double> it = price.iterator();
13         double total = 0;
14         double highest = 0;
15         System.out.println("\nProduct Prices:");
16         while (it.hasNext()) {
17             double p = it.next();
18             System.out.println(p);
19             total = total + p;
20             if (p > highest) {
21                 highest = p;
22             }
23         }
24         double average = total / price.size();
25         System.out.println("\nTotal Price = " + total);
26         System.out.println("Average Price = " + average);
27         System.out.println("Highest Price = " + highest);
28         sc.close();
29     }
30 }
                
```

Input

```

2.5
1
2
54
2.3
                
```

Output

```

1.0
2.0
54.0
Total Price = 596.5
Average Price = 59.65
                
```

Q2
Remove Even Numbers Using
Iterator
10 marks

Q3
Product Price Analysis Using
Iterator
10 marks

3/3 answered

Read the prices of 10 products from the user.
Traverse the list using an Iterator.
Calculate and display:
Total price of all products.
Average product price.
Highest product price.
Display all product prices using the Iterator.

Your Code

```
1 import java.util.*;
2 import java.util.*;
3 import java.util.*;
4 class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<Double> price = new ArrayList<>();
8         System.out.println("Enter 10 product prices:");
9         for (int i = 0; i < 10; i++) {
10             price.add(sc.nextDouble());
11         }
12         Iterator<Double> it = price.iterator();
13         double total = 0;
14         double highest = 0;
15         System.out.println("\nProduct Prices:");
16         while (it.hasNext()) {
17             double p = it.next();
18             System.out.println(p);
19             total = total + p;
20             if (p > highest) {
21                 highest = p;
22             }
23         }
24         double average = total / price.size();
25         System.out.println("\nTotal Price = " + total);
26         System.out.println("Average Price = " + average);
27         System.out.println("Highest Price = " + highest);
28         sc.close();
29     }
30 }
```

Input

25
1
2
54
23

Output

```
1.0
2.0
54.0
23.0
Total Price = 86.0
Average Price = 8.6
```